

33.5 Growth curves

A growth-curve model gives each subject a *curve of its own* — a straight line, a polynomial in time — the population curve being their average. How is the average curve estimated, and how is an individual's curve recovered from its own few observations? The second answer is the best linear unbiased predictor: a matrix-weighted average of the subject's own noisy curve and the precise population curve, weighted by signal against signal-plus-noise — the shrinkage of [Chapter 27](#) by a different road.

33.5.1 Two formulations

Let each of N subjects be observed at the same m times, and collect the data in an $m \times N$ matrix $\mathbf{Y}$ whose column i is subject i 's profile. Let $\mathbf{X}$ be the $m \times p$ matrix of a chosen basis in time, say $[\mathbf{1}, \mathbf{t}]$ for a straight line.

The **Potthoff–Roy** formulation (Potthoff and Roy 1964) writes

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X} \mathbf{\Xi} \mathbf{A}^\top, \quad \text{Cov}(\text{column } i) = \mathbf{\Sigma}, \text{ columns independent}, \quad (33.5.1)$$

where $\mathbf{A}$ is an $N \times k$ matrix of between-subject covariates (a column of ones if the subjects form one group, group indicators if not) and $\mathbf{\Xi}$ is the $p \times k$ matrix of curve coefficients, with $\mathbf{\Sigma}$ unstructured. It is a multivariate regression with a design on *both* sides: $\mathbf{X}$ acts on the times and $\mathbf{A}$ on the subjects.

The **random-coefficient** formulation gives each subject its own coefficient vector,

$$\mathbf{y}_i = \mathbf{X}(\boldsymbol{\beta} + \mathbf{u}_i) + \boldsymbol{\varepsilon}_i, \quad \mathbf{u}_i \sim (\mathbf{0}, \mathbf{D}) \text{ independent of } \boldsymbol{\varepsilon}_i \sim (\mathbf{0}, \sigma^2 \mathbf{I}_m), \quad (33.5.2)$$

with $\mathbf{u}_1, \dots, \mathbf{u}_N$ uncorrelated. This is the linear mixed model (Definition 32.2.1) with $\mathbf{Z}_i = \mathbf{X}$, $\mathbf{G} = \mathbf{D}$ and $\mathbf{R} = \sigma^2 \mathbf{I}$, and the implied within-subject covariance is

$$\mathbf{\Sigma} = \mathbf{X} \mathbf{D} \mathbf{X}^\top + \sigma^2 \mathbf{I}_m, \quad (33.5.3)$$

a covariance with $p(p+1)/2 + 1$ parameters instead of the $m(m+1)/2$ of [Eq. 33.5.1](#) — four, for a straight line, whatever m is. The two describe the same *mean* structure: with one group of subjects, the random-coefficient model is [Eq. 33.5.1](#) with $\mathbf{A} = \mathbf{1}_N$ and $\mathbf{\Sigma}$ restricted to [Eq. 33.5.3](#). That restriction gives $\mathcal{C}(\mathbf{\Sigma} \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, so the population curve is ordinary least squares and needs no covariance parameter ([Proposition 33.5.1\(a\)](#) below); with $\mathbf{\Sigma}$ unstructured the inclusion fails unless $p = m$, and the efficient estimate is a genuine generalized least squares fit.

33.5.2 Estimation

Proposition 33.5.1. The balanced random-coefficient growth model

Assume [Eq. 33.5.2](#) with a common $\mathbf{X}$ of full column rank $p \leq m$, and $\mathbf{D}$ positive definite. Write $\mathbf{K} = \mathbf{X}^\top \mathbf{X}$, $\mathbf{C} = \sigma^2 \mathbf{K}^{-1}$, $\mathbf{S} = \mathbf{D} + \mathbf{C}$, and let $\mathbf{b}_i = \mathbf{K}^{-1} \mathbf{X}^\top \mathbf{y}_i$ be subject i 's own least squares coefficient vector.

- (a) $\mathcal{C}(\mathbf{\Sigma} \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$, so ordinary and generalized least squares coincide, and

$$\hat{\boldsymbol{\beta}} = \mathbf{K}^{-1} \mathbf{X}^\top \bar{\mathbf{y}} = \bar{\mathbf{b}}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \frac{\mathbf{D} + \sigma^2 \mathbf{K}^{-1}}{N} = \frac{\mathbf{S}}{N}, \quad (33.5.4)$$

exactly, for every N . No covariance parameter is needed to estimate the population curve.

- (b) The best linear unbiased predictor of $\mathbf{u}_i$ (Theorem 32.3.1, which is Theorem 14.2.8 applied coordinatewise with $\mathbf{x}_0 = \mathbf{0}$) is

$$\hat{\mathbf{u}}_i = \mathbf{D} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \mathbf{X} \hat{\boldsymbol{\beta}}) = \mathbf{W} (\mathbf{b}_i - \hat{\boldsymbol{\beta}}), \quad \mathbf{W} = \mathbf{D} \mathbf{S}^{-1}, \quad (33.5.5)$$

with $\mathbf{W} = \mathbf{D}(\mathbf{D} + \mathbf{C})^{-1}$ and $\text{Cov}(\hat{\mathbf{u}}_i - \mathbf{u}_i) = \mathbf{D} - (1 - 1/N) \mathbf{W} \mathbf{D}$. The predicted curve of subject i is $\mathbf{X}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i)$, with

$$\text{Cov}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i - \boldsymbol{\beta} - \mathbf{u}_i) = (\mathbf{D}^{-1} + \mathbf{C}^{-1})^{-1} + (\mathbf{I} - \mathbf{W}) \frac{\mathbf{C}}{N}. \quad (33.5.6)$$

- (c) The restricted maximum likelihood estimates of σ^2 and $\mathbf{D}$ are

$$\hat{\sigma}^2 = \frac{1}{N(m-p)} \sum_{i=1}^N \|\mathbf{y}_i - \mathbf{X} \mathbf{b}_i\|^2, \quad \hat{\mathbf{D}} = \mathbf{S}_b - \hat{\sigma}^2 \mathbf{K}^{-1}, \quad (33.5.7)$$

where $\mathbf{S}_b = (N-1)^{-1} \sum_i (\mathbf{b}_i - \bar{\mathbf{b}})(\mathbf{b}_i - \bar{\mathbf{b}})^\top$, provided $\hat{\mathbf{D}}$ is nonnegative definite.

Proof.

- (a) $\boldsymbol{\Sigma} \mathbf{X} = \mathbf{X} \mathbf{D} \mathbf{K} + \sigma^2 \mathbf{X} = \mathbf{X}(\mathbf{D} \mathbf{K} + \sigma^2 \mathbf{I})$, whose columns lie in $\mathcal{C}(\mathbf{X})$. Stacking the subjects, the model matrix is $\mathbf{1}_N \otimes \mathbf{X}$ and the covariance is $\mathbf{I}_N \otimes \boldsymbol{\Sigma}$, so $(\mathbf{I}_N \otimes \boldsymbol{\Sigma})(\mathbf{1}_N \otimes \mathbf{X}) = \mathbf{1}_N \otimes \boldsymbol{\Sigma} \mathbf{X}$ has columns in $\mathcal{C}(\mathbf{1}_N \otimes \mathbf{X})$, and Theorem 6.9.3 gives the equality of the two fits (this is Theorem 31.2.2 in the balanced case). The ordinary least squares estimate from the stacked data is $(N\mathbf{K})^{-1} \sum_i \mathbf{X}^\top \mathbf{y}_i = \mathbf{K}^{-1} \mathbf{X}^\top \bar{\mathbf{y}} = \bar{\mathbf{b}}$. Its covariance is $\mathbf{K}^{-1} \mathbf{X}^\top (\boldsymbol{\Sigma}/N) \mathbf{X} \mathbf{K}^{-1}$, and $\mathbf{K}^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} \mathbf{K}^{-1} = \mathbf{K}^{-1}(\mathbf{K} \mathbf{D} \mathbf{K} + \sigma^2 \mathbf{K}) \mathbf{K}^{-1} = \mathbf{D} + \sigma^2 \mathbf{K}^{-1}$.

- (b) The first expression is Theorem 32.3.1. For the second, Theorem 1.6.5 gives $\boldsymbol{\Sigma}^{-1} = \sigma^{-2} \{\mathbf{I} - \mathbf{X}(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{X}^\top\}$, so $\mathbf{D} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} = \sigma^{-2} \mathbf{D} \{\mathbf{I} - \mathbf{K}(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1}\} \mathbf{X}^\top$. The braced matrix is $\sigma^2 \mathbf{D}^{-1}(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1}$, so $\mathbf{D} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} = (\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{X}^\top$; applying this to $\mathbf{y}_i - \mathbf{X} \hat{\boldsymbol{\beta}}$ and inserting $\mathbf{K} \mathbf{K}^{-1}$ gives $\hat{\mathbf{u}}_i = (\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{K} (\mathbf{b}_i - \hat{\boldsymbol{\beta}})$, and $(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{K} = (\sigma^2 \mathbf{I} + \mathbf{D} \mathbf{K})^{-1} \mathbf{D} \mathbf{K} = \mathbf{D}(\sigma^2 \mathbf{K}^{-1} + \mathbf{D})^{-1} = \mathbf{W}$.

For the prediction error covariance, Theorem 32.3.1 gives $\text{Cov}(\hat{\mathbf{u}} - \mathbf{u}) = \mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{P} \mathbf{Z} \mathbf{G}$ with $\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1} \mathbf{X}_* (\mathbf{X}_*^\top \mathbf{V}^{-1} \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{V}^{-1}$ for the stacked model. Here $\mathbf{Z} = \mathbf{I}_N \otimes \mathbf{X}$, $\mathbf{G} = \mathbf{I}_N \otimes \mathbf{D}$, $\mathbf{X}_* = \mathbf{1}_N \otimes \mathbf{X}$ and $\mathbf{V} = \mathbf{I}_N \otimes \boldsymbol{\Sigma}$, so $\mathbf{Z}^\top \mathbf{P} \mathbf{Z} = (\mathbf{I}_N - N^{-1} \mathbf{1}_N \mathbf{1}_N^\top) \otimes (\mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X})$ and the i th diagonal block of $\mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{P} \mathbf{Z} \mathbf{G}$ is $\mathbf{D} - (1 - 1/N) \mathbf{D} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X} \mathbf{D} = \mathbf{D} - (1 - 1/N) \mathbf{W} \mathbf{D}$, using $\mathbf{D} \mathbf{X}^\top \boldsymbol{\Sigma}^{-1} \mathbf{X} = \mathbf{W}$ from the first part.

For Eq. 33.5.6, write $\mathbf{b}_i = \boldsymbol{\beta} + \mathbf{u}_i + \mathbf{c}_i$ with $\mathbf{c}_i = \mathbf{K}^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}_i \sim (\mathbf{0}, \mathbf{C})$ independent of $\mathbf{u}_i$, so $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \bar{\mathbf{u}} + \bar{\mathbf{c}}$ and $\hat{\mathbf{u}}_i = \mathbf{W}(\mathbf{b}_i - \hat{\boldsymbol{\beta}})$. Then $\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}_i) = \{\text{Cov}(\bar{\mathbf{b}}, \mathbf{b}_i) - \text{Cov}(\bar{\mathbf{b}}, \bar{\mathbf{b}})\} \mathbf{W}^\top = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{u}_i) = \mathbf{D}/N$, so

$$\text{Cov}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i - \boldsymbol{\beta} - \mathbf{u}_i) = \frac{\mathbf{S}}{N} + \mathbf{D} - \left(1 - \frac{1}{N}\right) \mathbf{W} \mathbf{D} - \frac{2\mathbf{D}}{N}.$$

Now $\mathbf{D} - \mathbf{W} \mathbf{D} = \mathbf{D} - \mathbf{D} \mathbf{S}^{-1} \mathbf{D} = \mathbf{D} \mathbf{S}^{-1} (\mathbf{S} - \mathbf{D}) = \mathbf{W} \mathbf{C}$ and $\mathbf{I} - \mathbf{W} = (\mathbf{S} - \mathbf{D}) \mathbf{S}^{-1} = \mathbf{C} \mathbf{S}^{-1}$, so the right-hand side is $\mathbf{W} \mathbf{C} + (\mathbf{S} - 2\mathbf{D} + \mathbf{D} \mathbf{S}^{-1} \mathbf{D})/N = \mathbf{W} \mathbf{C} + \mathbf{C} \mathbf{S}^{-1} \mathbf{C}/N = \mathbf{W} \mathbf{C} + (\mathbf{I} - \mathbf{W}) \mathbf{C}/N$. Finally $\mathbf{W} \mathbf{C} = \mathbf{D} \mathbf{S}^{-1} \mathbf{C} = \{\mathbf{C}^{-1} \mathbf{S} \mathbf{D}^{-1}\}^{-1} = (\mathbf{C}^{-1} + \mathbf{D}^{-1})^{-1}$, which is Eq. 33.5.6.

- (c) An error contrast is a linear function of the data with mean zero for every $\boldsymbol{\beta}$ (Definition 32.4.1). Let $\mathbf{U}$ be $m \times (m - p)$ with columns an orthonormal basis of $\mathcal{C}(\mathbf{X})^\perp$, and consider the $Nm - p$ functions

$$\mathbf{w}_i = \mathbf{U}^\top \mathbf{y}_i \quad (i = 1, \dots, N), \quad \mathbf{b}_i - \bar{\mathbf{b}} \quad (i = 1, \dots, N).$$

They are error contrasts, and a maximal set: the map $\mathbf{y}_i \mapsto (\mathbf{b}_i, \mathbf{w}_i)$ is a bijection, the $\mathbf{w}_i$ contribute $N(m - p)$ independent functions and the $\mathbf{b}_i - \bar{\mathbf{b}}$ contribute $(N - 1)p$, totalling $Nm - p$. Now $\mathbf{w}_i = \mathbf{U}^\top \boldsymbol{\varepsilon}_i \sim N_{m-p}(\mathbf{0}, \sigma^2 \mathbf{I})$ independently, while $\mathbf{b}_i \sim N_p(\boldsymbol{\beta}, \mathbf{S})$ independently, and $\text{Cov}(\mathbf{b}_i, \mathbf{w}_i) = \sigma^2 \mathbf{K}^{-1} \mathbf{X}^\top \mathbf{U} = \mathbf{0}$; the two blocks are independent, so the restricted likelihood factors as

$$L_R(\sigma^2, \mathbf{D}) = L_1(\sigma^2) \cdot L_2(\mathbf{S}), \quad \mathbf{S} = \mathbf{D} + \sigma^2 \mathbf{K}^{-1},$$

with L_1 the likelihood of $N(m - p)$ independent $N(0, \sigma^2)$ variables and L_2 that of $\mathbf{S}$ from the contrasts $\mathbf{b}_i - \bar{\mathbf{b}}$. Since $(\sigma^2, \mathbf{D}) \mapsto (\sigma^2, \mathbf{S})$ is a bijection onto $\{\mathbf{S} - \sigma^2 \mathbf{K}^{-1} \succeq \mathbf{0}\}$, the unconstrained maximizers are the restricted maximum likelihood estimates whenever they satisfy the constraint. Maximizing L_1 gives $\hat{\sigma}^2 = \sum_i \|\mathbf{y}_i - \mathbf{X} \mathbf{b}_i\|^2 / \{N(m - p)\}$, and maximizing L_2 , the restricted likelihood of a multivariate normal mean model with design $\mathbf{1}_N$, gives $\hat{\mathbf{S}} = \mathbf{S}_b$ (Theorem 32.4.2). Subtracting yields $\hat{\mathbf{D}}$. ■

Part (a) is a statement about the restriction Eq. 33.5.3, not about balance alone: the population curve is ordinary least squares on the *average profile*, which is also the average of the subjects' own curves, and nothing has to be estimated first. Potthoff and Roy's own proposal was not to estimate $\boldsymbol{\Sigma}$ at all: transforming each profile by a positive definite weight matrix reduces Eq. 33.5.1 to an ordinary multivariate analysis of variance, whose tests are exact for weights chosen without looking at the data. The efficient estimate came later, with Rao (1965); estimating $\boldsymbol{\Sigma}$ from the residual profiles and weighting by it is feasible generalized least squares (Definition 31.4.1).

Idea. $\mathbf{W} = \mathbf{D}(\mathbf{D} + \mathbf{C})^{-1}$ compares two covariances: $\mathbf{D}$, how much subjects really differ, and $\mathbf{C} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$, how noisily each subject's own curve is measured. When subjects differ a lot relative to the noise each keeps its own curve, $\mathbf{W} \approx \mathbf{I}$; when they hardly differ every subject is given the population curve, $\mathbf{W} \approx \mathbf{0}$. Between those the predictor borrows strength across subjects, coordinate by coordinate: an intercept may be barely shrunk while a slope is shrunk by a quarter.

Part (c) says that restricted maximum likelihood here is the obvious moment estimator: pool the within-subject residual sums of squares for σ^2 , and from the spread of the subjects' own coefficient vectors subtract the part that is estimation noise. The subtraction can produce a matrix that is not nonnegative definite — the growth-curve version of the negative variance component of Proposition 32.1.1 — and the estimate then sits on the boundary, where the closed form no longer applies.

33.5.3 Whose curve

Two prediction problems must be kept apart. For a subject **in the data** the predicted curve is $\mathbf{X}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i)$, with error covariance Eq. 33.5.6: the first term is what remains even with $\boldsymbol{\beta}$ known,

the second the price of estimating it. For a subject **not in the data** nothing is known but its membership of the population, so the best predictor is $\mathbf{X}\hat{\boldsymbol{\beta}}$ with error covariance $\mathbf{D} + \mathbf{S}/N$ — far larger, as it should be.

33.5.4 An example

Example. The investment trajectories of eleven firms

The panel of Example (Investment in eleven firms), seen a third time — after the random firm effect of [Chapter 32](#) and the covariance models of [Section 33.4](#) — now with a straight line in t (decades from the middle of the record) whose intercept *and slope* vary by firm. The within-firm residual standard deviation is $\hat{\sigma} = 0.2797$, and

$$\hat{\mathbf{D}} = \begin{pmatrix} 2.3635 & 0.0455 \\ 0.0455 & 0.0361 \end{pmatrix},$$

so firms' log-investment levels have standard deviation 1.5374 — enormous, the difference between General Motors and Diamond Match — and their slopes 0.1899 per decade, correlated 0.156 with the level. Maximizing the restricted likelihood numerically returns the same three values to four decimals, as [Proposition 33.5.1\(c\)](#) predicts.

The population curve has intercept 3.9109 and slope 0.5577 per decade, with standard errors 0.4639 and 0.0659 from [Eq. 33.5.4](#); the slope is 8.46 standard errors from zero. The shrinkage matrix is

$$\mathbf{W} = \begin{pmatrix} 0.9983 & 0.0016 \\ 0.0048 & 0.7496 \end{pmatrix},$$

so intercepts are left alone — with twenty observations per firm a firm's own level is known almost exactly — while slopes are pulled a quarter of the way towards the population slope. The firms' own slopes span 0.158 to 0.943, the predicted slopes 0.248 to 0.846, a standard deviation 0.755 times that of the raw slopes; the largest adjustment is IBM's, from 0.943 to 0.846. [Figure 33.5.1](#) shows the eleven firms with all three lines.

For the last year of the record, $t = 0.95$, the predicted value on a firm's own curve has standard error 0.1102, against 1.6461 for a firm outside the data — fifteen times larger, and simulation from the fitted model confirms both, at 0.1103 and 1.6603. The interval for a new firm, 1.214 to 7.667 on the log scale, is nearly useless, and rightly so: without knowing a firm's size, its investment cannot be predicted within three orders of magnitude.

```
import numpy as np
import statsmodels.api as sm

df = sm.datasets.grunfeld.load_pandas().data
firms = sorted(df["firm"].unique())
years = np.sort(df["year"].unique())
N, m = len(firms), len(years)
t = (years - years.mean()) / 10           # decades from the middle of
    the record
X = np.column_stack([np.ones(m), t])      # the same design for every
    firm
p = X.shape[1]
Y = np.column_stack([np.log(df[df["firm"] == f].sort_values("year")["invest"]).
    to_numpy()
                    for f in firms])      # m x N: one column per firm
```

[Run this code online](#)

```
XtXinv = np.linalg.inv(X.T @ X)
B = XtXinv @ X.T @ Y                                # p x N: each firm's own least
    squares line
beta_hat = B.mean(axis=1)                             # the average of the firm lines
E = Y - X @ B                                         # within-firm residuals
sigma2 = np.sum(E**2) / (N * (m - p))                 # pooled within-firm variance
Sb = np.cov(B, ddof=1)                               # spread of the firm lines
D_hat = Sb - sigma2 * XtXinv                         # ... minus what estimation
    noise adds
print("beta_hat", beta_hat.round(4), " sigma", round(np.sqrt(sigma2), 4))
print("D_hat\n", D_hat.round(4))
print("implied sd of firm intercepts and slopes:", np.sqrt(np.diag(D_hat)).round(4))
```

[Run this code online](#)

```
Sigma = X @ D_hat @ X.T + sigma2 * np.eye(m)
cov_beta = (D_hat + sigma2 * XtXinv) / N              # exact covariance of beta_hat
W = D_hat @ np.linalg.inv(D_hat + sigma2 * XtXinv)    # the shrinkage matrix
B_blup = beta_hat[:, None] + W @ (B - beta_hat[:, None])
u_direct = D_hat @ X.T @ np.linalg.solve(Sigma, Y - (X @ beta_hat)[:, None])
print("shrinkage matrix\n", W.round(4))
print("largest difference from the direct formula:",
      np.abs(B_blup - beta_hat[:, None] - u_direct).max())
```

[Run this code online](#)

```
C = sigma2 * XtXinv                                  # noise in a firm's own line
V_known = np.linalg.inv(np.linalg.inv(D_hat) + np.linalg.inv(C)) \
    + (np.eye(p) - W) @ C / N                        # error of the predicted firm
    curve
V_new = D_hat + cov_beta                             # error for a firm not in the
    data
x_end = np.array([1.0, t[-1]])
se_curve_known = np.sqrt(x_end @ V_known @ x_end)
se_curve_new = np.sqrt(x_end @ V_new @ x_end)
pred_new = x_end @ beta_hat
print(f"1954 curve: se for a firm in the data {se_curve_known:.4f}, "
      f"for a new firm {se_curve_new:.4f}")
```

[Run this code online](#)

Remark (Beyond the balanced straight line). Three extensions matter in practice and none changes the ideas. Subjects observed at different times, or a different number of times: [Eq. 33.5.4](#) fails, the fit becomes genuine generalized least squares with $\Sigma_i = \mathbf{X}_i \mathbf{D} \mathbf{X}_i^\top + \sigma^2 \mathbf{I}$, and Henderson's equations (Theorem 32.3.2) do the computing. Between-subject covariates: replace β by $\Xi \mathbf{a}_i$ as in the formulation [Eq. 33.5.1](#). Serial correlation on top of the random coefficients: take $\mathbf{R}$ autoregressive, as in [Section 33.4](#). Beyond all of these lies the question whether a polynomial in time is the right shape at all: [Chapter 42](#) and [Chapter 43](#).

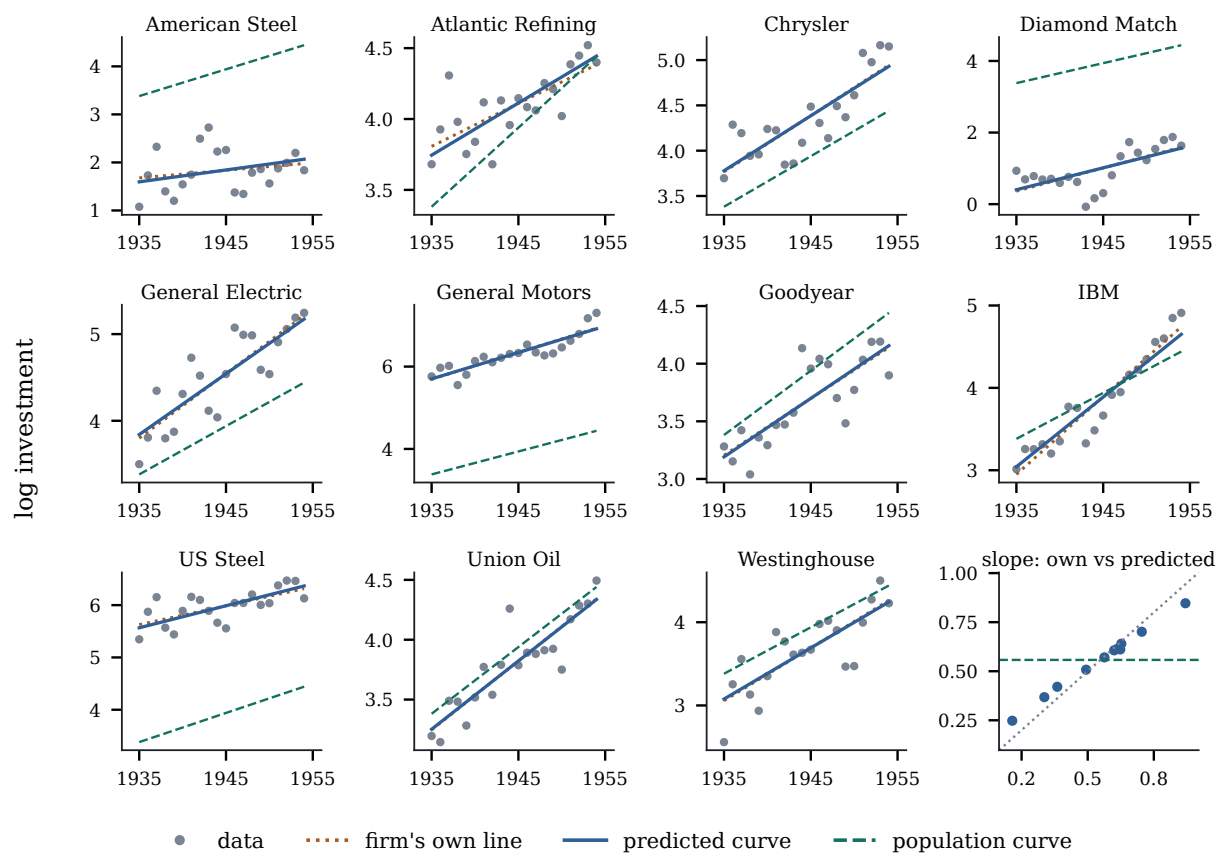


Figure 33.5.1: Log gross investment of eleven firms, 1935–1954, with each firm's own least squares line (dotted), its predicted curve $\mathbf{X}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i)$ (solid) and the population curve $\mathbf{X}\hat{\boldsymbol{\beta}}$ (dashed). The last panel plots own slope against predicted slope.

33.5.5 Exercises

33.5.5.1 A. Check your understanding

Exercise. A1

Take $p = 1$, so $\mathbf{X} = \mathbf{1}_m$ and each subject has a single random level. Show that $\mathbf{W}$ reduces to $\sigma_u^2/(\sigma_u^2 + \sigma^2/m)$ and that the predictor is the shrunk mean $\hat{\boldsymbol{\beta}} + \mathbf{W}(\bar{y}_i - \hat{\boldsymbol{\beta}})$. What happens as $m \rightarrow \infty$, and as $\sigma_u^2 \rightarrow 0$?

Solution. With $\mathbf{X} = \mathbf{1}_m$, $\mathbf{K} = m$, $\mathbf{C} = \sigma^2/m$ and $\mathbf{D} = \sigma_u^2$, so $\mathbf{W} = \sigma_u^2/(\sigma_u^2 + \sigma^2/m)$ and $\mathbf{b}_i = \bar{y}_i$. As $m \rightarrow \infty$ the noise in $\bar{y}_i$ vanishes, $\mathbf{W} \rightarrow 1$ and no shrinkage occurs; as $\sigma_u^2 \rightarrow 0$, $\mathbf{W} \rightarrow 0$ and every subject is given the common mean, correctly, since the subjects are then identical.

Exercise. A2

The predicted curves are less spread out than the true ones. Is this a bias? Compute $E(\hat{\mathbf{u}}_i)$ and $\text{Cov}(\hat{\mathbf{u}}_i)$ and compare the latter with $\mathbf{D}$.

33.5.5.2 B. Practice

Exercise. B1

Suppose the fixed effects use a quadratic in time, $\mathbf{X} = [\mathbf{1}, \mathbf{t}, \mathbf{t}^2]$, but the random effects only a line, $\mathbf{Z} = [\mathbf{1}, \mathbf{t}]$. Show that $\mathcal{C}(\boldsymbol{\Sigma}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ still holds, and identify what changes in $\text{Cov}(\hat{\boldsymbol{\beta}})$. What if instead the random effects used the quadratic and the fixed effects only the line?

Exercise. B2

Let subject i be observed at its own times, giving $\mathbf{X}_i$ of full column rank p , and let $\mathbf{b}_i = (\mathbf{X}_i^\top \mathbf{X}_i)^{-1} \mathbf{X}_i^\top \mathbf{y}_i$. Show that $\mathbf{b}_i \sim (\boldsymbol{\beta}, \mathbf{D} + \sigma^2(\mathbf{X}_i^\top \mathbf{X}_i)^{-1})$ independently over i , and deduce that the generalized least squares estimate from the $\mathbf{b}_i$ is $\hat{\boldsymbol{\beta}} = (\sum_i \mathbf{W}_i)^{-1} \sum_i \mathbf{W}_i \mathbf{b}_i$ with $\mathbf{W}_i = \{\mathbf{D} + \sigma^2(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}\}^{-1}$, a precision-weighted average of the subjects' own curves. Recover [Eq. 33.5.4](#) as the balanced case.

Solution. $\mathbf{b}_i = \boldsymbol{\beta} + \mathbf{u}_i + (\mathbf{X}_i^\top \mathbf{X}_i)^{-1} \mathbf{X}_i^\top \boldsymbol{\varepsilon}_i$ is a sum of independent terms with covariances $\mathbf{D}$ and $\sigma^2(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}$, and different subjects are independent. The $\mathbf{b}_i$ are therefore independent with common mean $\boldsymbol{\beta}$ and covariances $\mathbf{W}_i^{-1}$, so Corollary 7.2.3 applied to the stacked $\mathbf{b}_i$ with design $\mathbf{1}_N \otimes \mathbf{I}_p$ gives the weighted average. When every $\mathbf{X}_i = \mathbf{X}$ the weights cancel, leaving $\hat{\boldsymbol{\beta}} = \bar{\mathbf{b}}$ with covariance $\mathbf{W}_1^{-1}/N$, which is [Eq. 33.5.4](#).

33.5.5.3 C. Going deeper

Exercise. C1

Simulate data for which $\hat{\mathbf{D}}$ of Eq. 33.5.7 is not nonnegative definite. Describe what the restricted maximum likelihood estimate does instead, relate it to the boundary problem of Proposition 32.5.1, and explain why the two-stage estimate is nevertheless unbiased for $\mathbf{D}$.